

Dynamical Systems and Optimal Control

Content

- Examples of optimization problems
- Calculus of variations, classical examples
- Optimal control: examples; mathematical formalization;
- Controllability of linear systems
- Bang-bang principle
- Time-optimal controls of linear systems
- Pontryagin Maximum Principle

Literature

- L. Evans, An Introduction to Mathematical Optimal Control Theory, lectures, 2024, University of California, Berkeley
- D. Liberzone, Calculus of variations and optimal control theory, Princeton University Press, 2012
- Wendell Fleming, Raymond Rishel, Deterministic and stochastic optimal control, Springer, 1975